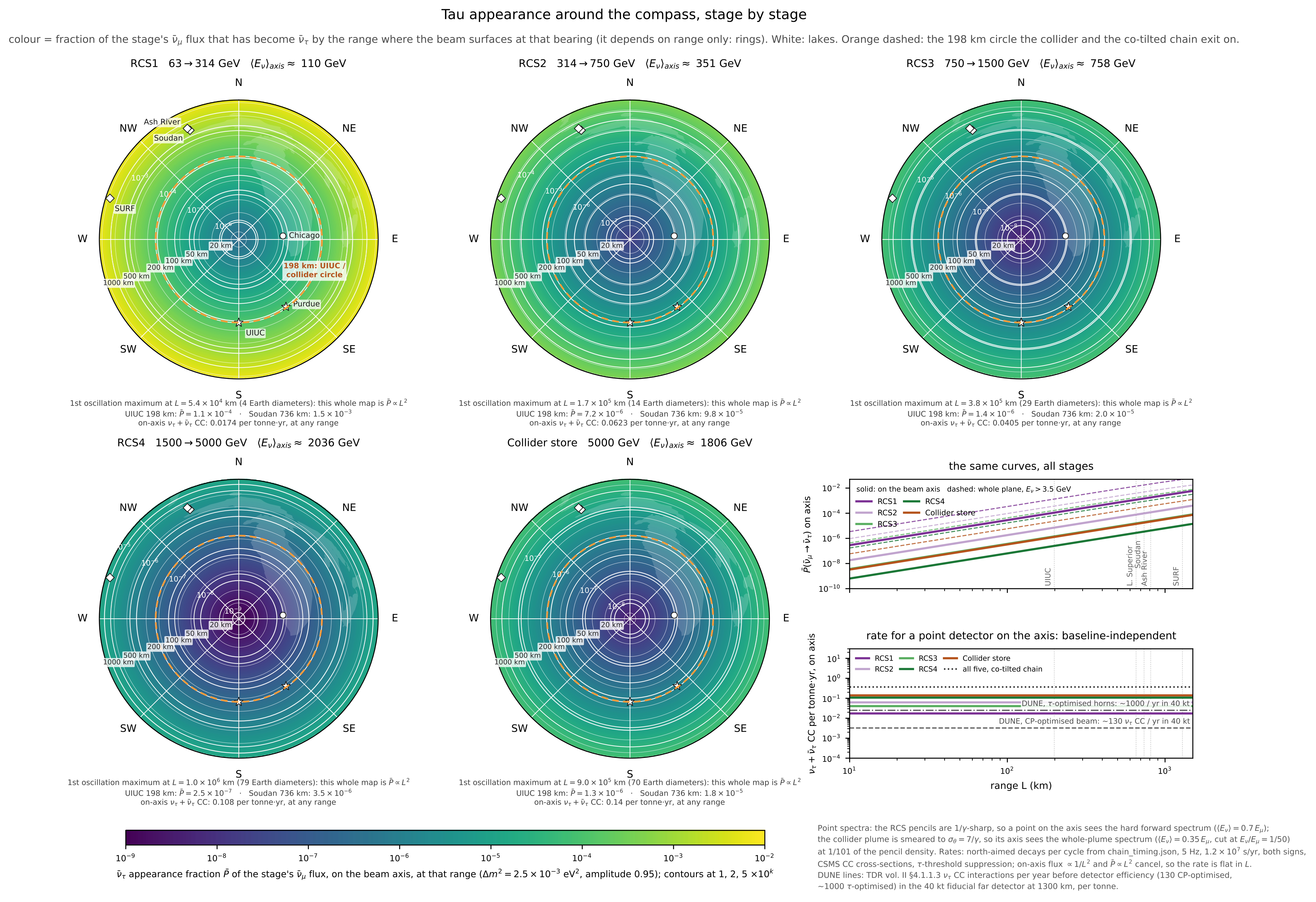

Tau appearance around the compass, stage by stage
colour = fraction of the stage's $\bar\nu_\mu$ flux that has become $\bar\nu_\tau$ by the range where the beam surfaces at that bearing (it depends on range only: rings). White: lakes. Orange dashed: the 198 km circle the collider and the co-tilted chain exit on.
RCS1 63 → 314 GeV $\langle E_\nu\rangle_{axis} \approx$ 110 GeV
N
NE
E
SE
S
SW
W
NW
Ash River
Soudan
SURF
Chicago
Purdue
UIUC
198 km: UIUC / collider circle
20 km
50 km
100 km
200 km
500 km
1000 km
1st oscillation maximum at $L = 5.4 \times 10^4$ km (4 Earth diameters): this whole map is $\bar P \propto L^2$
UIUC 198 km: $\bar P = 1.1 \times 10^{-4}$ · Soudan 736 km: 1.5×10^{-3}
on-axis $\nu_\tau + \bar\nu_\tau$ CC: 0.0174 per tonne·yr, at any range
RCS2 314 → 750 GeV $\langle E_\nu\rangle_{axis} \approx$ 351 GeV
1st oscillation maximum at $L = 1.7 \times 10^5$ km (14 Earth diameters): this whole map is $\bar P \propto L^2$
UIUC 198 km: $\bar P = 7.2 \times 10^{-6}$ · Soudan 736 km: 9.8×10^{-5}
on-axis $\nu_\tau + \bar\nu_\tau$ CC: 0.0623 per tonne·yr, at any range
RCS3 750 → 1500 GeV $\langle E_\nu\rangle_{axis} \approx$ 758 GeV
1st oscillation maximum at $L = 3.8 \times 10^5$ km (29 Earth diameters): this whole map is $\bar P \propto L^2$
UIUC 198 km: $\bar P = 1.4 \times 10^{-6}$ · Soudan 736 km: 2.0×10^{-5}
on-axis $\nu_\tau + \bar\nu_\tau$ CC: 0.0405 per tonne·yr, at any range
RCS4 1500 → 5000 GeV $\langle E_\nu\rangle_{axis} \approx$ 2036 GeV
1st oscillation maximum at $L = 1.0 \times 10^6$ km (79 Earth diameters): this whole map is $\bar P \propto L^2$
UIUC 198 km: $\bar P = 2.5 \times 10^{-7}$ · Soudan 736 km: 3.5×10^{-6}
on-axis $\nu_\tau + \bar\nu_\tau$ CC: 0.108 per tonne·yr, at any range
Collider store 5000 GeV $\langle E_\nu\rangle_{axis} \approx$ 1806 GeV
1st oscillation maximum at $L = 9.0 \times 10^5$ km (70 Earth diameters): this whole map is $\bar P \propto L^2$
UIUC 198 km: $\bar P = 1.3 \times 10^{-6}$ · Soudan 736 km: 1.8×10^{-5}
on-axis $\nu_\tau + \bar\nu_\tau$ CC: 0.14 per tonne·yr, at any range
the same curves, all stages
solid: on the beam axis dashed: whole plane, $E_\nu > 3.5$ GeV
RCS1
RCS2
RCS3
RCS4
Collider store
$\bar P(\bar\nu_\mu \to \bar\nu_\tau)$ on axis
UIUC
L. Superior
Soudan
Ash River
SURF
rate for a point detector on the axis: baseline-independent
all five, co-tilted chain
DUNE, τ-optimised horns: ~1000 / yr in 40 kt
DUNE, CP-optimised beam: ~130 ν_τ CC / yr in 40 kt
$\nu_\tau + \bar\nu_\tau$ CC per tonne·yr, on axis
range L (km)
$\bar\nu_\tau$ appearance fraction $\bar P$ of the stage's $\bar\nu_\mu$ flux, on the beam axis, at that range ($\Delta m^2 = 2.5 \times 10^{-3}$ eV^2, amplitude 0.95); contours at 1, 2, 5 $\times 10^k$
Point spectra: the RCS pencils are $1/\gamma$-sharp, so a point on the axis sees the hard forward spectrum ($\langle E_\nu\rangle = 0.7\,E_\mu$);
the collider plume is smeared to $\sigma_\theta = 7/\gamma$, so its axis sees the whole-plume spectrum ($\langle E_\nu\rangle = 0.35\,E_\mu$, cut at $E_\nu/E_\mu = 1/50$)
at 1/101 of the pencil density. Rates: north-aimed decays per cycle from chain_timing.json, 5 Hz, 1.2×10^7 s/yr, both signs,
CSMS CC cross-sections, τ-threshold suppression; on-axis flux $\propto 1/L^2$ and $\bar P \propto L^2$ cancel, so the rate is flat in L.
DUNE lines: TDR vol. II §4.1.1.3 ν_τ CC interactions per year before detector efficiency (130 CP-optimised,
~1000 τ-optimised) in the 40 kt fiducial far detector at 1300 km, per tonne.